

AI MOST IMPORTANT

TREND ANALYSIS (Year-wise Insight)

1. Highly Repetitive Core Areas (Asked Every Year)

These topics appear **every year in some form** (2 marks / 10 marks / conceptual or analytical):

Topic	Evidence
Definition & scope of AI	All 3 papers – Section A
Intelligent Agents (types, architecture, PEAS)	2022–23, 2023–24, 2024–25
Search strategies (uninformed, informed, heuristics, local search)	Every year
Logic & reasoning (FOPL, unification, resolution, chaining)	Every year
Multi-Agent Systems (communication, negotiation, bargaining)	Every year
NLP / Language models / Speech recognition	Every year
Information Retrieval vs Information Extraction	Every year

 **Conclusion:** These are **must-prepare** topics.

2. Shift in Question Nature (Important Observation)

Year	Focus
2022–23	Algorithmic, problem solving (A*, Alpha-Beta, Water Jug, CSP)
2023–24	Conceptual + ethics + language models
2024–25	Analytical + future-oriented (Quantum AI, NLP future, trust, reputation)

Trend:

AKTU is moving from **pure algorithms** → **applied + analytical AI concepts**.

3. Bloom's Taxonomy Trend

- Early papers: **K1–K2 (define, explain)**
- Latest paper: **K4–K6 (evaluate, examine, analyze future impact)**

➡ Expect “**explain with justification / advantages / future impact**” type questions.

★ **MOST IMPORTANT QUESTIONS (HIGH PROBABILITY)**

● **SECTION A (2 MARKS) – VERY HIGH HIT RATE**

Prepare crisp definitions for:

1. Define Artificial Intelligence and its objectives
2. Types of AI problems
3. Uninformed vs Informed search
4. Heuristic function
5. Local search
6. Unification in Prolog
7. Resolution in logic programming
8. Forward chaining vs Backward chaining
9. Components of intelligent agent architecture
10. Information Retrieval vs Information Extraction

➡ **80–90% chances** similar questions appear.

● **SECTION B (10 MARKS) – CORE SCORING AREA**

🔥 **MOST REPEATED (Prepare First)**

1. **Uninformed & Informed Search Strategies with examples**
2. **Role of heuristics in problem solving**
3. **First Order Predicate Logic (FOPL) & Prolog**
4. **Sensors and Effectors in Intelligent Agents**
5. **Applications of AI & impact on industries**

🔄 **FREQUENTLY ALTERNATING**

6. PEAS representation of an agent

7. Constraint Satisfaction Problems (CSP)
 8. Local search algorithms
 9. Language models & pre-trained models
 10. Ethical issues in AI
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● SECTION C (10 MARKS) – DEEP UNDERSTANDING REQUIRED

★ VERY HIGH PROBABILITY

1. Reactive vs Proactive / Deliberative agents (with applications)
 2. AI problem-solving approach using search & heuristics
 3. Future of AI in NLP (limitations & solutions)
 4. Local search algorithms with example
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● LONG ANSWERS (Sections 4–7) – GAME CHANGERS

Prepare **any ONE** from each group:

◆ Search & Problem Solving

- A* Algorithm – completeness & optimality
- Backtracking search in CSPs
- Problem solving methods in AI vs traditional methods

◆ Knowledge & Reasoning

- FOPL vs Propositional Logic
- Forward vs Backward chaining
- Ontological engineering

◆ Multi-Agent Systems (VERY HOT TOPIC)

- Negotiation & Bargaining
- Communication paradigms
- Trust & reputation in MAS

◆ NLP & IR

- Speech recognition steps & importance

- Language models in NLP
 - Information Retrieval in search engines
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FINAL EXAM STRATEGY (SMART PREP)

- If time is less:
 - 👉 Prepare **Agents + Search + Logic + MAS + NLP**
 - If aiming 70+ marks:
 - 👉 Add **Ethics, CSP, Ontology, IR**
 - **Diagrams required:**
Agents architecture, A*, CSP, NLP pipeline
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SUMMARY

- ~65–70% syllabus repeats every year
 - Focus is shifting toward **analysis & application**
 - Multi-Agent Systems + NLP are **trend leaders**
 - This question list is **AKTU-oriented, not generic**
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